x_{ij} presents in any cell means that an assignment is made their.In such cases $x_{ij} = 1$.

The assignment model can be written in LPP as follows:

$$\text{Minimize } Z = \sum_{i=1}^{m} \sum_{j=1}^{n} C_{ij}\, x_{ij}$$

Subject to the constrains

$$\sum_{i=1}^{n} x_{ij} = 1,\ j = 1,2,.....n$$

$$\sum_{j=1}^{n} x_{ij} = 1,\ i = 1,2,.....n \quad \text{and} \quad x_{ij} = 0$$

(or) 1 for all i,j

Note

The optimum assignment schedule remains unaltered if we add or subtract a constant from all the elements of the row or column of the assignment cost matrix.

Note

If for an assignment problem all $C_{ij} > 0$ then an assignment schedule (x_{ij}) which satisfies $\sum C_{ij}\, x_{ij} = 0$ must be optimal.

10.2.2 Solution of assignment problems (Hungarian Method)

First check whether the number of rows is equal to the numbers of columns, if it is so, the assignment problem is said to be balanced.

Step :1 Choose the least element in each row and subtract it from all the elements of that row.

Step :2 Choose the least element in each column and subtract it from all the elements of that column. Step 2 has to be performed from the table obtained in step 1.

Step:3 Check whether there is atleast one zero in each row and each column and make an assignment as follows.

(i) Examine the rows successively until a row with exactly one zero is found. Mark that zero by □, that means an assignment is made there . Cross $(\times)$ all other zeros in its column. Continue this until all the rows have been examined.

(ii) Examine the columns successively until a columns with exactly one zero is found. Mark that zero by □, that means an assignment is made there . Cross $(\times)$ all other zeros in its row. Continue this until all the columns have been examined.

Step :4 If each row and each column contains exactly one assignment, then the solution is optimal.

DO YOU KNOW?

Hungarian method provides optimum assignment schedule in an assignment problem.

Example 10.7

Solve the following assignment problem. Cell values represent cost of assigning job A, B, C and D to the machines I, II, III and IV.

		machines			
		I	II	III	IV
jobs	A	10	12	19	11
	B	5	10	7	8
	C	12	14	13	11
	D	8	15	11	9

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